

Muhammad Umar Khan

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SUMMARY

M.S. Computer Science candidate specializing in full-stack and AI-driven systems. Built async processing pipelines, LLM-integrated APIs, and JWT-secured backends across Java, C#, and Python using Spring Boot, ASP.NET Core, React, PostgreSQL, Docker, and AWS.

EDUCATION

Wayne State University | Detroit, MI, USA

Jan. 2025 – Aug. 2026

M.S. in Computer Science | GPA: 3.92

University of Windsor | Windsor, ON, Canada

Sep. 2020 – Apr. 2024

B.S. in Computer Science with Software Engineering Specialization

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript/TypeScript, SQL

Frameworks: Spring Boot, Spring Security, ASP.NET Core, FastAPI, Node.js, React, Tailwind CSS

Databases: PostgreSQL, MongoDB

Tools & DevOps: Git, Docker, AWS (Elastic Beanstalk, RDS, S3, CloudFront), Railway, Vercel, REST APIs, JWT Authentication, JUnit, Redis, Celery, Gemini API

EXPERIENCE

University of Windsor | Windsor, ON, Canada

Apr. 2023 – Nov. 2024

Research Assistant

- Engineered an image segmentation pipeline from scratch using KMeans clustering and Euclidean distance calculations to partition images into superpixel regions with configurable overlap.
- Reimplemented hybrid CNN+GNN architecture from published research, constructing SLIC-based superpixel graph construction and a coupled model for image classification.
- Implemented and benchmarked CNN (TensorFlow/sklearn) and GCN (PyTorch Geometric) models, achieving 98% on MNIST and 81% on Cora, comparing classification accuracy across architectures.

University of Windsor | Windsor, ON, Canada

Sep. 2022 – Apr. 2024

Teaching Assistant

- Assisted 100+ students through code reviews and debugging sessions across Data Structures, OOP, and Database Management Systems.
- Led weekly lab sessions of up to 30 students on Java, data structures, and algorithm design.

PROJECTS

AutoOps AI – [\[Live\]](#) | ASP.NET Core, C#, React, TypeScript, FastAPI, PostgreSQL, Redis, Celery, Docker, Railway, Vercel

- Built a polyglot AI document processing platform — ASP.NET Core (C#) REST API with JWT auth and EF Core over PostgreSQL, backed by a Python Gemini 2.5 Flash microservice over PostgreSQL.
- Designed a Redis/Celery pipeline processing PDFs, DOCX, and TXT through 4 async stages with real-time SSE progress and AI-generated summaries streamed to a React frontend.
- Containerized 5 services with Docker Compose; deployed to Railway and Vercel with environment-driven config.

Khan's Blog Platform – [\[Live\]](#) | Java, Spring Boot, React, TypeScript, PostgreSQL, AWS

- Deployed a full-stack blog platform on AWS (Elastic Beanstalk, RDS PostgreSQL, S3, CloudFront), routing all API traffic through CloudFront as a reverse proxy to enforce unified HTTPS delivery across 3 distributed services.
- Implemented JWT authentication with RBAC (Admin/User), resource-level permissions, and a React frontend with debounced search, category/tag filtering, dark/light mode, and real-time notifications across 10+ REST endpoints.

AdaptIQ – [\[Live\]](#) | React, FastAPI, Python, JavaScript, Google Gemini API, Railway

- Built AI resume optimizer using a single-pass LLM pipeline for keyword extraction, reducing Gemini API calls by 50%.
- Deployed full-stack to Railway with JWT authentication middleware securing all API routes and a synonym-aware ATS scoring engine resolving 30+ tech skill aliases to eliminate false negatives in gap analysis.
- Designed a 5-metric ATS scoring engine (parsing ability, keyword injection, coverage, quantification rate, job match) that programmatically scans tailored resume text and surfaces actionable improvement insights per generation.